

Lecture 7: Fresnel relations for reflection and transmission at a boundary

- **Explicit derivation for p (TM) polarization**
- **Key features of the results for both s (TE) and p (TM) polarizations**

Monochromatic Plane Waves:

$$\frac{\omega}{k} = c = \frac{c_0}{n}$$

$$\begin{aligned}\vec{E}(\vec{r}, t) &= \text{Re}(\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t + \varphi_0)}) \\ &= \frac{\vec{E}_0 e^{i\varphi_0}}{2} e^{i(\vec{k} \cdot \vec{r} - \omega t)} + \frac{\vec{E}_0 e^{-i\varphi_0}}{2} e^{-i(\vec{k} \cdot \vec{r} - \omega t)} \\ &\equiv \vec{E}_0^{(+)} e^{i(\vec{k} \cdot \vec{r} - \omega t)} + \vec{E}_0^{(-)} e^{-i(\vec{k} \cdot \vec{r} - \omega t)} \\ &= \vec{E}_0^{(+)} e^{i(\vec{k} \cdot \vec{r} - \omega t)} + c.c.\end{aligned}$$

Intensity of light

is time average

*over many optical
cycles*

$$\langle |\vec{S}| \rangle = \frac{\langle |\vec{E} \times \vec{B}| \rangle}{\mu} = \frac{\langle |\vec{E}|^2 \rangle}{\mu c} = \frac{\langle 2\vec{E}^{(+)} \cdot \vec{E}^{(-)} \rangle}{\mu c} = \frac{2 |\vec{E}_0^{(+)}|^2}{\mu c}$$

Applies very generally

*For a monochromatic
plane wave*

Reflection and transmission at an interface: Maxwell's equations give boundary conditions

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{\ell} = 0$$

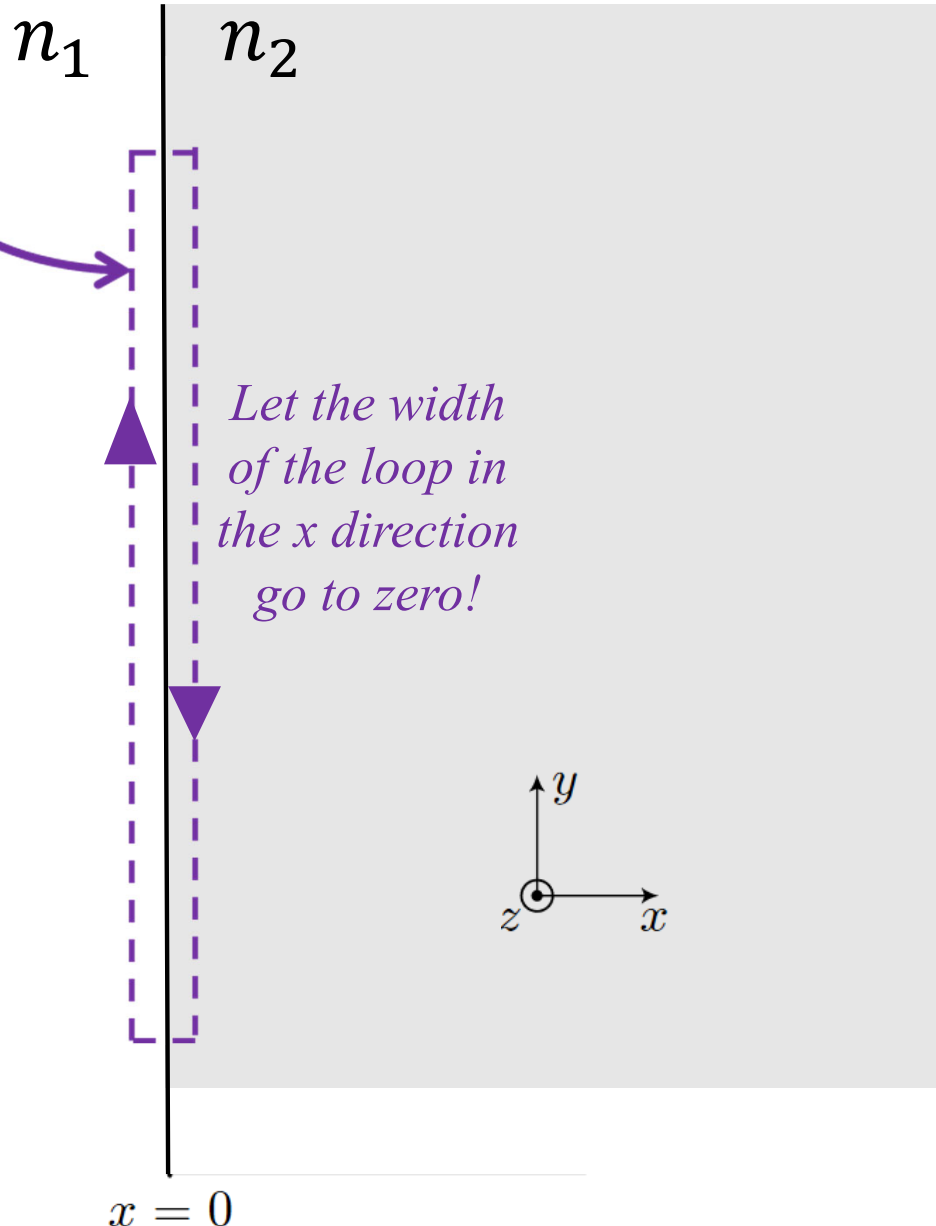
around this loop

$$\vec{\nabla} \times \vec{H} = \vec{j}_{free} + \frac{\partial \vec{D}}{\partial t}$$

$$\oint \vec{H} \cdot d\vec{\ell} = 0$$

around this loop

*Tangential components of $\vec{E}$ and $\vec{H}$
are continuous across the boundary.*



Starting calculation for P (or TM) polarization:

For each of incident, reflected, and transmitted wave:

$$\vec{E}(x, y, z, t) = \vec{E}_0^{(+)} e^{i(k_x x + k_y y + k_z z)} e^{-i\omega t} + c.c.$$

At interface $x = 0$.

By choice of coords $k_z = 0$.

$$\vec{E}(0, y, z, t) = \vec{E}_0^{(+)} e^{ik_y y} e^{-i\omega t} + c.c.$$

$$\vec{H}(0, y, z, t) = \vec{H}_0^{(+)} e^{ik_y y} e^{-i\omega t} + c.c.$$

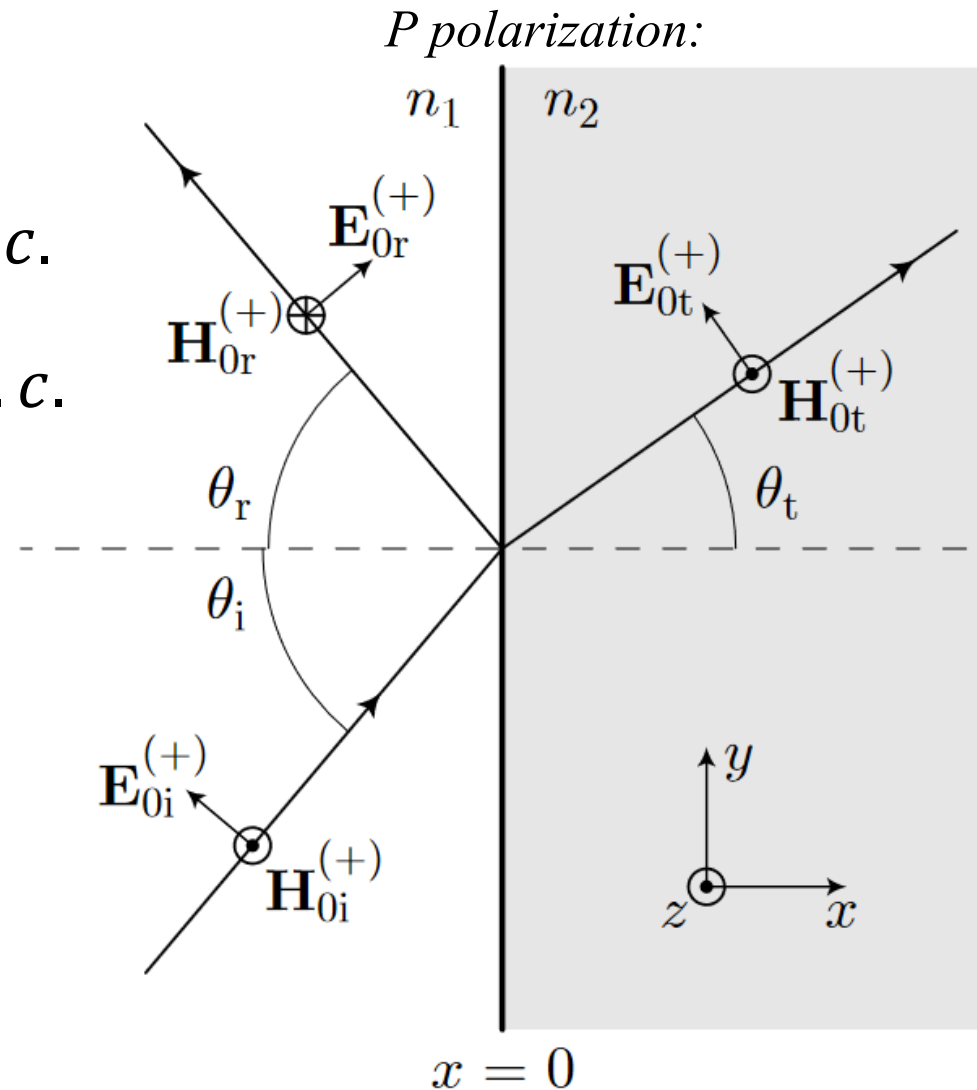
$$H_0^{(+)} = \frac{E_0^{(+)}}{\mu_0 c} = \frac{E_0^{(+)} n}{\mu_0 c_0}$$

Impose continuity of H_z :

$$\omega_i = \omega_r = \omega_t$$

$$\theta_r = \theta_i$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$



Fresnel relations for p (TM) polarization:

$$H_{0i}^{(+)} e^{ik_i \sin \theta_i y} e^{-i\omega_i t} - H_{0r}^{(+)} e^{ik_r \sin \theta_r y} e^{-i\omega_r t} = H_{0t}^{(+)} e^{ik_t \sin \theta_t y} e^{-i\omega_t t}$$

$$E_{0i}^{(+)} \cos \theta_i e^{ik_i \sin \theta_i y} e^{-i\omega_i t} + E_{0r}^{(+)} \cos \theta_r e^{ik_r \sin \theta_r y} e^{-i\omega_r t} = E_{0t}^{(+)} \cos \theta_t e^{ik_t \sin \theta_t y} e^{-i\omega_t t}$$

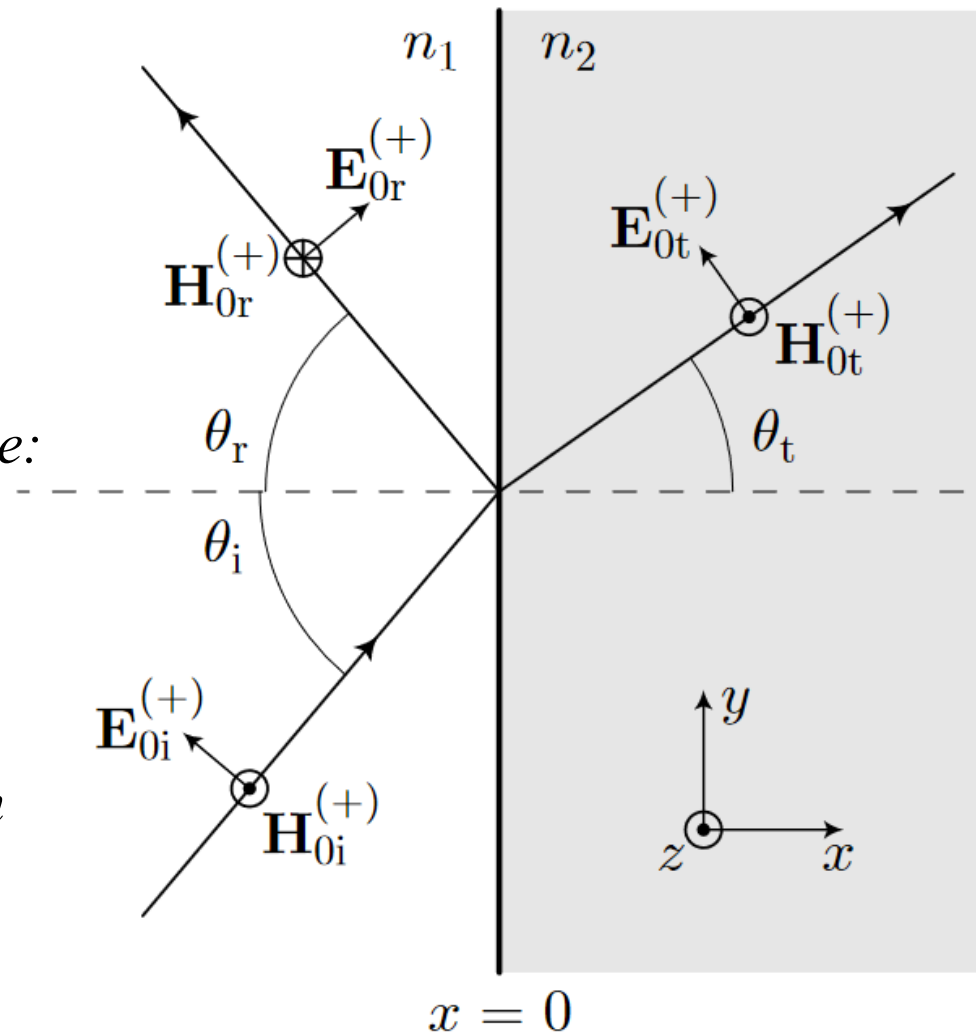
$$r \equiv \frac{E_{or}^{(+)}}{E_{oi}^{(+)}} = \frac{n_1 \cos \theta_t - n_2 \cos \theta_i}{n_1 \cos \theta_t + n_2 \cos \theta_i}$$

Reflected frac. of intensity incident on interface:

$$R = |r|^2 = \left| \frac{n_1 \cos \theta_t - n_2 \cos \theta_i}{n_1 \cos \theta_t + n_2 \cos \theta_i} \right|^2$$

Brewster's angle: no reflection of p pol. when

$$\theta_i + \theta_t = \frac{\pi}{2} \quad \tan \theta_i = \frac{n_2}{n_1}$$



Fresnel relations for s (TE) polarization:

Done in detail in Steck 9.1.

Some things the same:

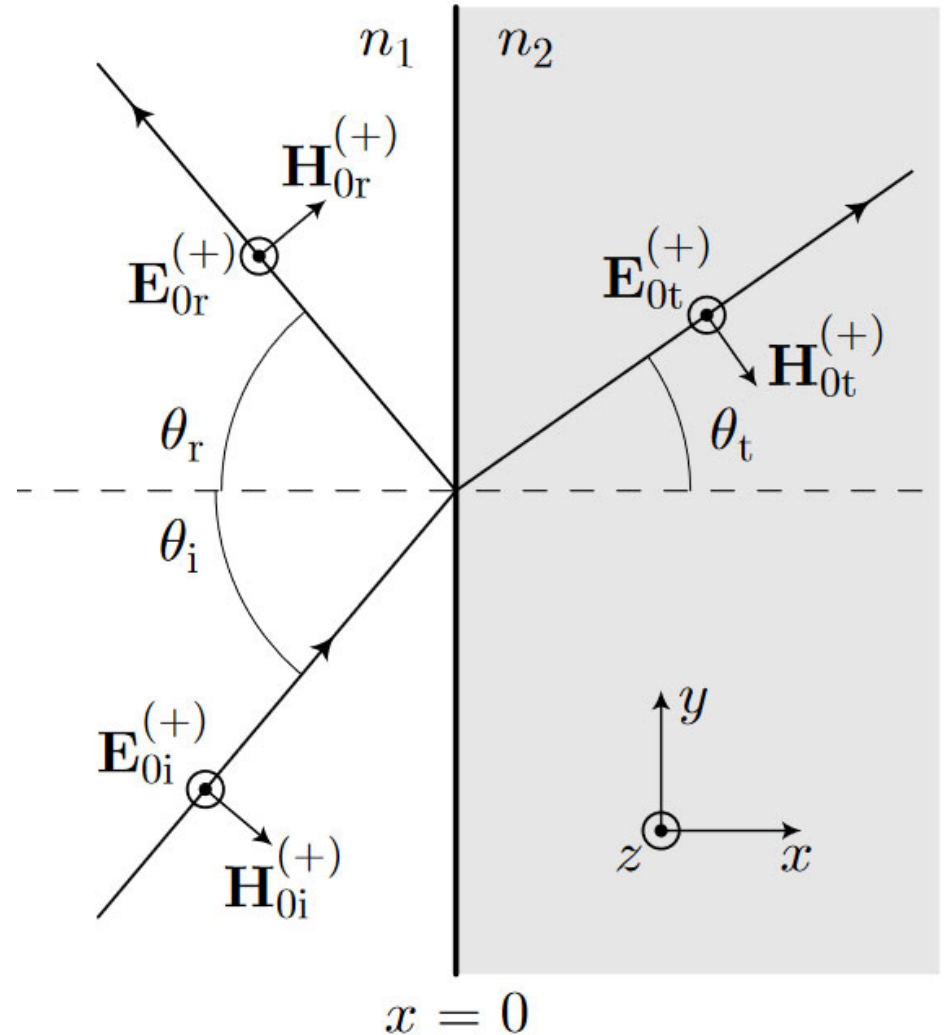
$$\omega_i = \omega_r = \omega_t$$

$$\theta_r = \theta_i$$

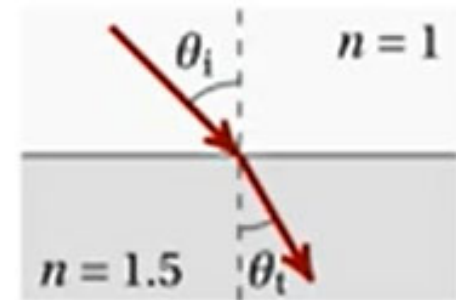
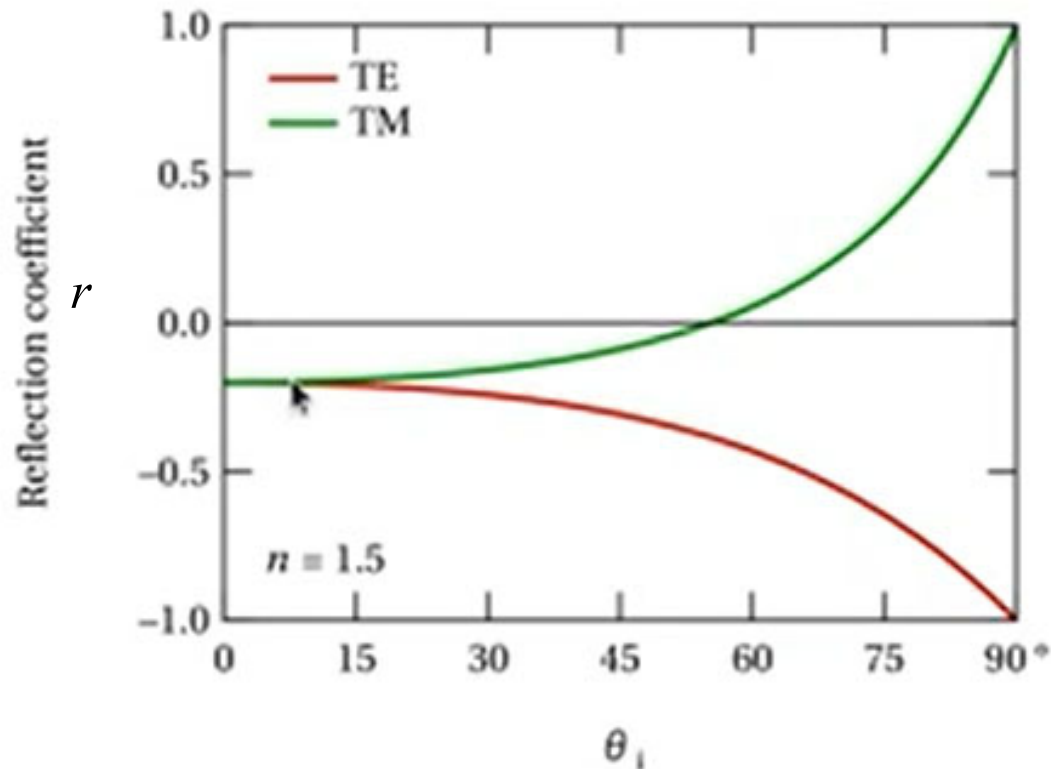
$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

*Reflection and transmission coefficients different.
(Thus reflectivity different.)*

S polarization:



Example reflection coefficients (air to glass):



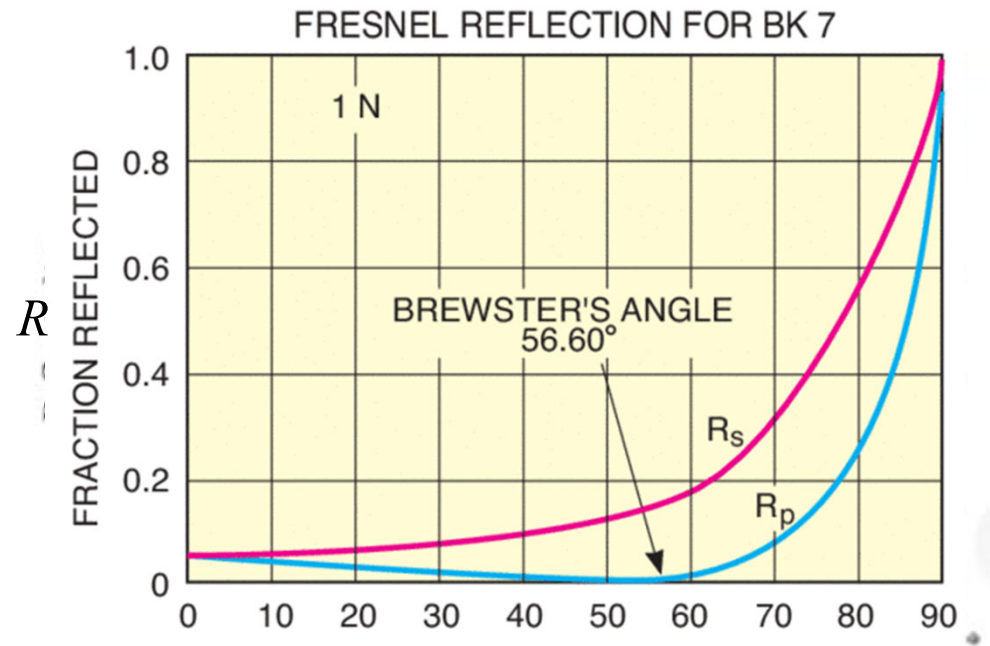
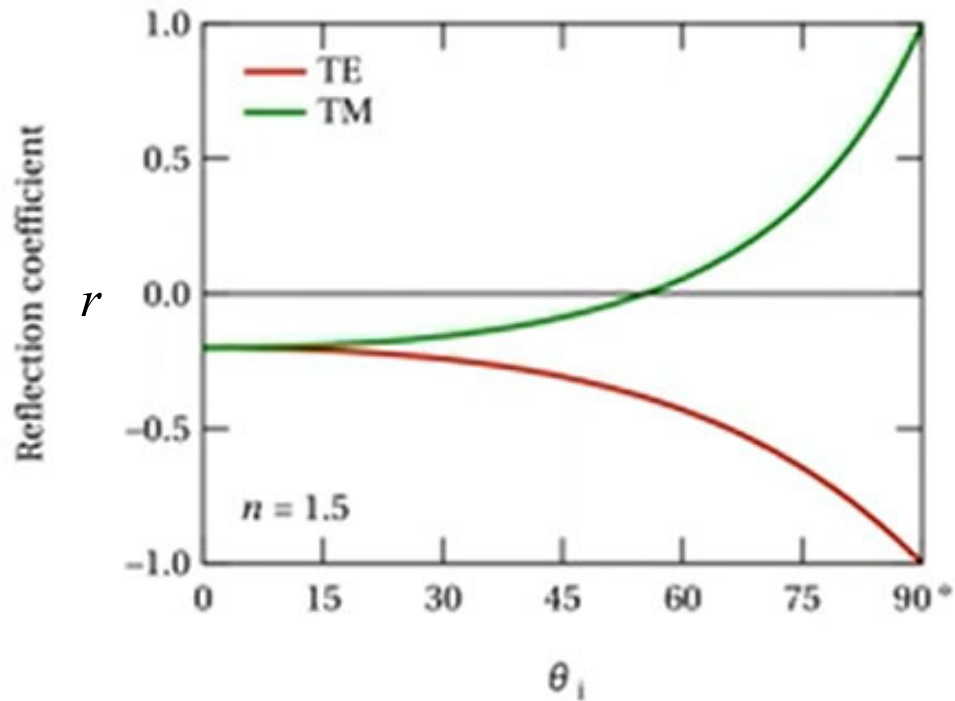
Reflected and transmitted fractions of intensity incident on interface:

$$R = |r|^2$$

$$T = 1 - R$$

Reflectivity

Air to glass:



Glass to air:

